

OPSC6311_POE_PART_1_PLANNING_AND_DESIGN

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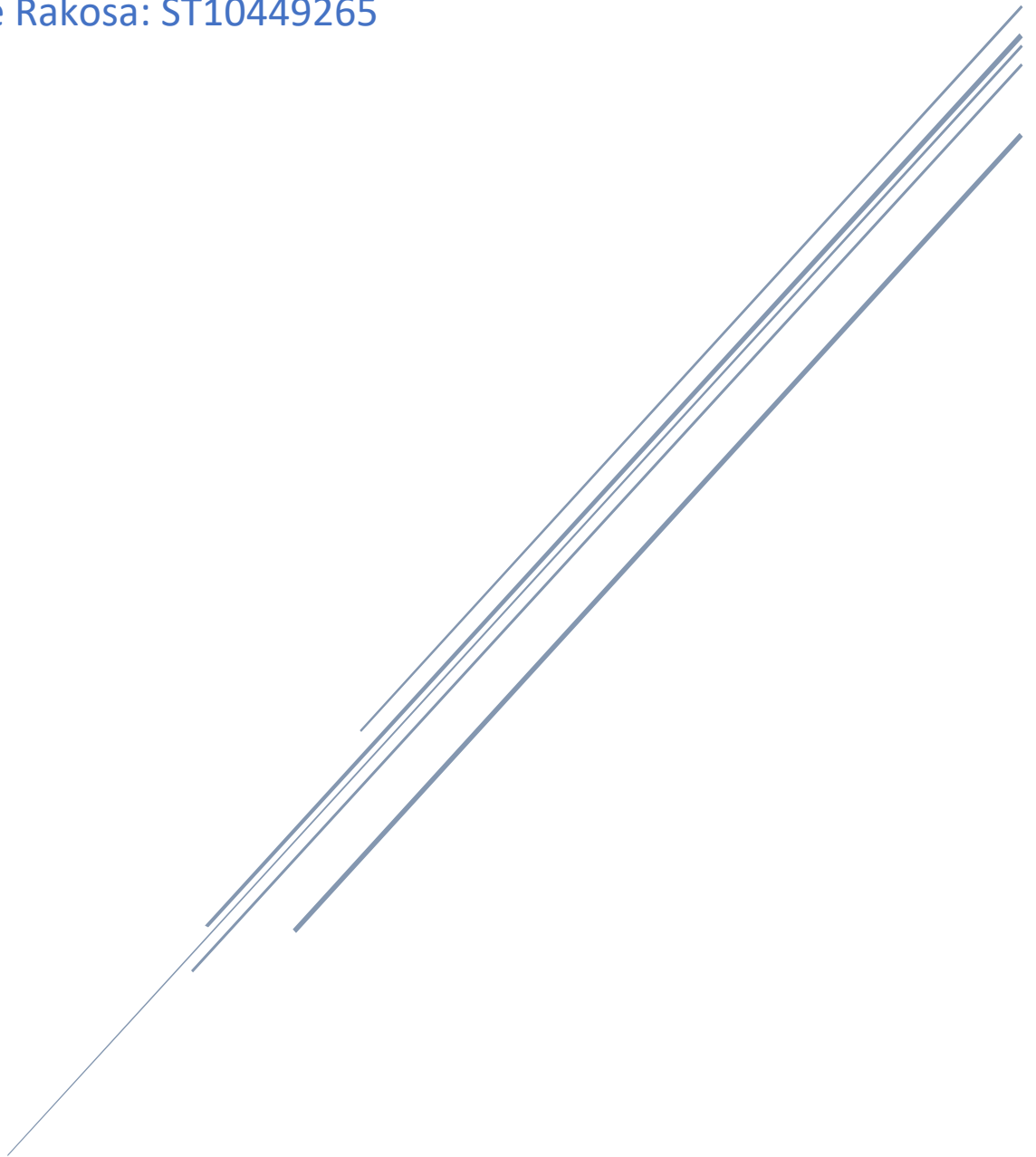


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Planning and Design: BudgetWise

Introduction

This document explains the design of BudgetWise app, a personal budgeting mobile app. The design was directly influenced by three well-known financial tools, which are YNAB (You Need A Budget), Good Budget, and Credit Karma. YNAB focuses on the need of proactive, goal-oriented budgeting (WalletHub, 2025), Good Budget shows how the outdated wallet method may be updated for digital cooperation and responsible spending (WalletHub, 2025), and Credit Karma focused on the advantages of combining budgeting with a full understanding of financial health (John Marshall Bank, 2026), which includes credit ratings and personalized insights.

The BudgetWise app combines these features into a single, unified software. Plus, the design combines motivational aspects, such as badges and awards, to encourage regular engagement and reinforce good financial habits. This document includes the app's basic functionality, user interface, navigation, and a detailed project plan with milestones to ensure that the development process is clearly defined and aligned with the research findings.

App Overview

BudgetWise is a budgeting tool app that uses its simple and visually appealing interface to help users in tracking costs, creating budgets, and achieving savings objectives. The app's name accurately represents its primary goal which is to aid users in making sound financial decisions.

The first design includes a neat, minimalist interface with a bottom navigation bar for easy access to all important elements.

Icon:



(Hassan, 2025).

Innovative features include:

- Smart Budgeting which is inspired by YNAB inspired encourages proactive financial planning by assigning a purpose to each unit of money spent (WalletHub, 2026).
- The Envelope Budgeting System which is inspired by Good-Budget which allows users to allocate each month's earnings to virtual "wallets" for specific spending categories, reducing overspending and encouraging smart purchasing (WalletHub, 2026).
- The holistic financial dashboard, which is inspired by Credit-Karma, offers users snapshots of their overall financial wellness, including trends and optional credit score insights for (for example, a "budget health score") that link spending habits to long-term goals (John Marshall Bank, 2026).
- Photo Receipts which Attach photos, such as receipts, to expense entries for convenient reference and record-keeping.
- Gamification of the app so that the software rewards users with badges like "Budget Master" for achieving monthly goals and "Consistent Logger" for tracking spending daily for a week, boosting long-term commitment.

Detailed Requirements

Functional Requirements:

- Users should be able to register a new account with a username and password and then log in to access their personal information.
- Users must be able to add, make changes, and delete custom expense categories.
- Users must be able to create an expense entry that includes the amount, date, description, and category. They can optionally include a photo (receipt) with their entry.
- Users should be able to specify a monthly total budget objective as well as individual spending limitations for each category (envelope).
- Users must be able to see a list of all expenses filtered by a user-selectable date period. If a receipt photo was attached, tapping the entry will display it.
- Users should be able to see the total amount spent in each category over a user-selectable date range.
- All the data, which includes users, categories, expenses, budgets, and images, must be saved in a local database.
- Users should be able to view a graph of daily spending over a user-selectable time period. To notice trends, the graph should clearly reflect category spending.
- A dashboard should show the user's progress toward the current month's budget targets. Overspending categories must be visually indicated for example, in red.
- The software must award badges to users who meet predefined goals, such as remaining under budget for three months in a row or tracking costs for seven days in a row.

Non-Functional Requirements:

- Any sensitive data, including user passwords, must be protected.
- The user interface must smoothly adjust to various screen sizes for example, for phones and tablets.
- Database searches and graph rendering should take no more than two seconds on average hardware.

User Interface Design & Navigation

The application's user interface has a basic design with a consistent color palette (green, blue, and white) to convey trust and clarity. The bottom navigation allows you to reach the five main screens with a single press.

Screen 1: Dashboard (Home).

- Should Displays the user's entire balance (revenue minus expenses), a monthly budget progress bar, a list of overspending notifications, and the most recent badges received. Inspired by Credit Karma, a "Budget Health Score" provides a quick assessment of financial wellness and with bottom navigation bar.

Screen 2: Transactions.

- Provides an organized list of expenses for a specific calendar range. Each entry should shows the amount, category, date, and a camera icon if a receipt photo is available. A floating action button ("+") brings up the "Add Expense" panel.
- Navigation: hitting a cost opens a detail view with a photo of the receipt; hitting "+" opens a modal form where you may add the amount, date, description, category, and, optionally, take or select a photo.

Screen 3 displays budgets.

- Users may choose a monthly overall budget as well as individual category restrictions (envelopes). Visual progress bars comparing current spending to each limit. Overspent envelopes are shown in red. The "Fill Envelopes" option allows users to distribute their available money at the beginning of each month.
- Navigation: Tapping a category brings up a screen where you can adjust its spending limit.

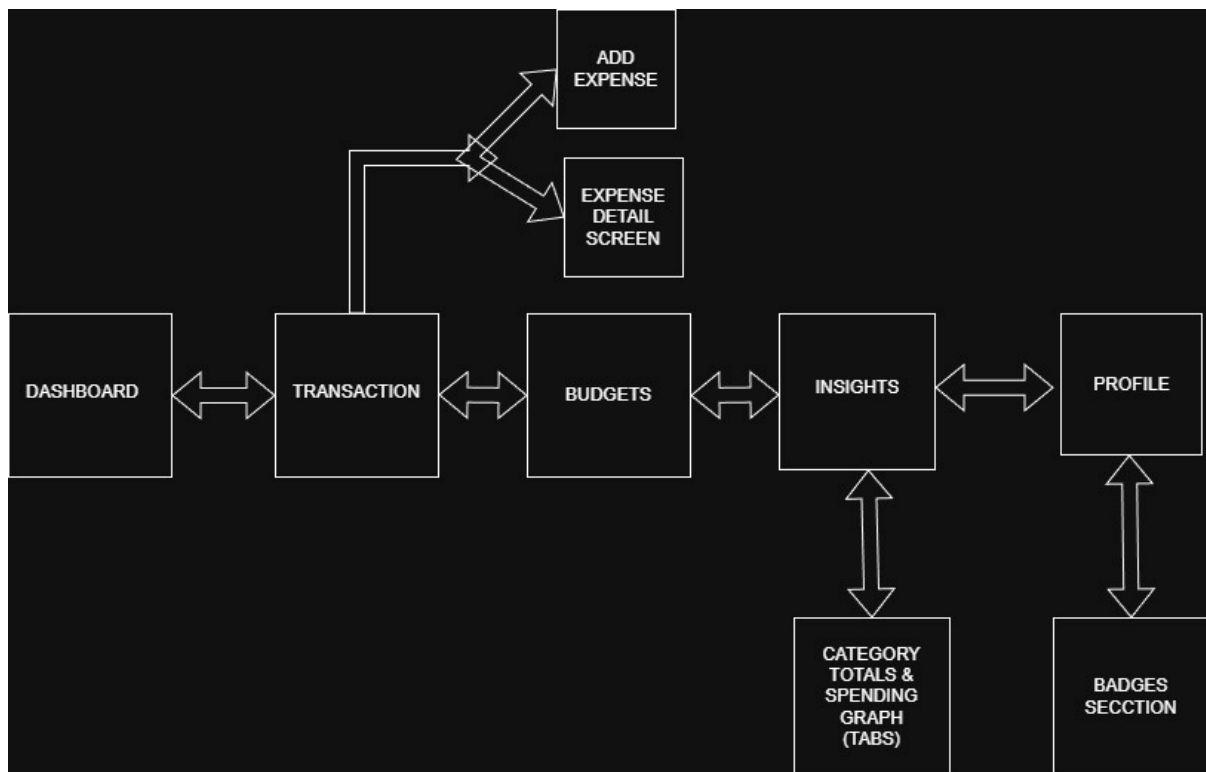
Screen 4: Insights.

- should Include two tabs:
- Category Totals: Displays a list of categories and their total spending for the given period.
- Spending Graph: Displays a daily line/bar graph of spending over the specified period, color-coded by category.
- Navigation: A date picker at the top allows users to modify the period. Tapping a graph segment leads to the related transactions.

Screen 5: Profile.

- should Include user account settings, a logout option, and a "achievement" area that will lists all received gamification awards/badges along with descriptions of how they were obtained.
- Navigation: Tapping a badge should displays its entire requirements and date earned.

Navigation Diagram:



(draw.io, 2025).

Project Plan (Gantt Chart)

The project will span six weeks; here are all the activities in order:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1									PROJECT PLAN TIMELINE					
2	Task/ Milestone													
3		WEEK 1	WEEK 2	WEEK 3	WEEK 4	WEEK 5	WEEK 6	WEEK 7	WEEK 8	WEEK 9	WEEK 10	WEEK 11	WEEK 12	WEEK 13
4	Requirements & Design													
5	Finalize requirements													
6	Create wireframes & UI mockups													
7	Design database schema													
8														
9	Core Development													
10	Set up project structure & local DB													
11	Implement user registration/login													
12	Implement category management													
13	Implement expense entry													
14	Implement photo attachment & storage													
15														
16	Advanced Features													
17	Implement spending graph													
18	Progress dashboard & alerts													
19	Gamification (badges rewards)													
20														
21	Testing & Bug Fixing													
22	Unit & Bug Fixing													
23	UI/UX Testing & Bug Fixing													

(Microsoft, 2024).

Conclusion

The BudgetWise app is meant to be a practical, interesting, and research-based personal finance tool. The software meets user needs completely by merging YNAB's proactive budgeting method with Good Budget's envelope-based clarity (The CFO Club, 2026), Credit Karma's detailed financial information, and creative gaming (John Marshall Bank, 2026). The detailed checklist will ensure that all required features from registration and spending logging to budget tracking and graphs are completely implemented. The project plan, with its detailed task breakdown as well as dedicated testing phase, will provide a realistic roadmap for timely delivery of a high-quality, stable application. Finally, BudgetWise app is intended to help people take control of their finances in a motivating and sustainable way.

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